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Power semiconductor device and preparation method thereof

Abstract

Disclosed are a power semiconductor device and a preparation method thereof, which belongs to the field of power semiconductor devices. By introducing a punch-through triode structure for electron extraction in a drift region, the frontside hole injection efficiency is reduced, and hole currents from emitters are converted into electron drift currents without significantly increasing on-state voltage drop. In addition, by changing the density and topography of a frontside trench, the adjustment of punch-through area and position is achieved, which in turn changes the electron extraction and frontside hole injection efficiency. The disclosed method increases the design flexibility and design dimension of the device.

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Background/Summary

FIELD OF TECHNOLOGY

(1) The present disclosure belongs to the field of power semiconductor devices, relates to an insulated gate bipolar transistor (IGBT), and in particular relates to a reverse conducting-insulated gate bipolar transistor (RC-IGBT) device and a preparation method thereof.

BACKGROUND

(2) Reverse conducting-insulated gate bipolar transistor (RC-IGBT) device, as an insulated gate bipolar transistor (IGBT) and a free-wheeling diode (FWD) are integrated in the same cell, has advantages such as increased power density, reduced parasitic inductance, and increased chip area utilization of the device compared to a manner of encapsulated integration of discrete IGBT and FWD into a single device by wire bonding. The RC-IGBT, by virtue of the advantages of high integration, has gained popularity in soft switching circuits, especially induction heating applications. But it cannot be further generalized in hard switching applications due to the limitations such as switching frequency and switching losses. Excessive reverse recovery loss of

FWD is one of the power consumption issues that urgently needs to be solved for RC-IGBT.

(3) From backside structures, RC-IGBT cells may consider heavily doped P.sup.+ collector region portion as the IGBT cell and the heavily doped N.sup.+ collector region portion as the FWD cell, as shown in FIG. 1. Due to difference in local doping on the back of the RC-IGBT cells, there is snap back when the RC-IGBT cell is forwardly conducted in an IGBT mode, thus the device is prone to local current gathering during operating, which degrades the reliability of the device. In the prior art, the snap back is generally eliminated by adopting a structure that a plurality of IGBT cells are connected in parallel with one FWD cell, while the metal-oxide-semiconductor structure of IGBT (also called MOS structure) is adopted for the frontside structure. To ensure the high breakdown voltage and suppress latch-up effects, the P region in the IGBT frontside structure generally employs a highly doped P-type region. The highly doped P region-type region, as an anode for free-wheeling conduction of the FWD cell, is excessive in hole injection efficiency, which results in excessive reverse recovery loss of the FWD cell during reverse recovery and limits the reduction of overall switching power loss of the device.

(4) To reduce the reverse recovery loss of the FWD cell, there is a solution employing the local lifetime control around the frontside region in the prior art. The specific practice is as follows: near the frontside PN junction, defects are introduced into the local region in a manner of H.sup.+ or He.sup.2+ particle implantation to enhance the carrier recombination process. When the FWD cell is forwardly conducted, the hole injection efficiency can be reduced, and then the number of carries in the drift region before reverse recovery is reduced to achieve the “low” frontside and “high” backside distribution of the carriers. Such carrier distribution can achieve soft and fast characteristics during the reverse recovery of the FWD cell and can reduce the reverse recovery time and reverse recovery charge, which reduces the reverse recovery loss of the RC-IGBT. However, the above technical solution has the problems of large leakage current at high temperature and increased conduction loss under the IGBT mode.

(5) Another solution in the prior art is to independently design the frontside structures of the IGBT cell and FWD cell of the RC-IGBT. The main feature is that the frontside hole injection efficiency of the FWD cell can be individually optimized while the impact on the frontside hole injection efficiency of the IGBT cell is reduced. The specific practice is to employ the existing discrete FWD device frontside structure, such as a self-adjusting P emitter efficiency diode (SPEED) structure. The principle of the technical solution is to introduce a Schottky junction formed by the metal-P-type semiconductor into the frontside P region, which can increase the barrier height for frontside hole injection without changing the process of transporting electrons to the frontside emitter. Such technical solution can change the barrier height of the Schottky junction by changing the doping concentration of the frontside P-type region and work functions of emitter metal, thus adjusting the hole injection efficiency. However, the parameter design using this technology is limited by the RC-IGBT process, leading to limitation on the reduction of reverse recovery loss. A method of controlling by IGBT cell gates is also proposed in the industry, the IGBT cell gates are turned on before the reverse recovery of FWD cell. Then the number of carriers during reverse recovery is reduced by way of deriving the frontside electron currents. However, such solution makes the gate drive control complicated and requires additional device for commutation detection, which is not conducive to controlling the system reliability and cost.

(6) In accordance with the problems in the existing reverse recovery loss reduction techniques for RC-IGBTs, there is a need to provide an innovative device structure to control leakage current and reduce the reverse recovery loss of RC-IGBT without the performance degradation of the IGBT cell, which can satisfy the high reliability requirement for the entire RC-IGBT device in harsh application environments.

SUMMARY

(7) To solve the problems above, the present disclosure provides a power semiconductor device. The device includes a collector electrode located at the bottom. A heavily doped first conductivity

type cathode region and a heavily doped second conductivity type collector region are arranged on the collector electrode in a staggered manner. A first conductivity type buffer layer is located on the heavily doped first conductivity type cathode region and a heavily doped collector region. A first conductivity type drift region is located on the first conductivity type buffer layer and an emitter electrode is located at the top of the device. The emitter electrode is spaced from the first conductivity type drift region. More than one frontside structure units are arranged on the first conductivity type drift region. Each of the frontside structure units includes: a trench gate structure, a second conductivity type base region adjacent to the trench gate structure, a lightly doped second conductivity type base region located on one side of the second conductivity type base region to isolate a trench auxiliary gate structure from the second conductivity type base region, the trench auxiliary gate structure, and a floating FP region located at the bottom of the trench auxiliary gate structure. The trench gate structure includes a first gate dielectric layer and a first gate electrode. The floating FP region is separated from the second conductivity type base region and the lightly doped second conductivity type base region. An upper surface of the second conductivity type base region is provided with a first heavily doped first conductivity type emitter region and a heavily doped second conductivity type emitter region. An upper surface of the lightly doped second conductivity type base region is provided with a second heavily doped first conductivity type emitter region. And the first heavily doped first conductivity type emitter region is separated from the second heavily doped first conductivity type emitter region by means of the heavily doped second conductivity type emitter region. An upper surface of the trench gate structure is provided with an insulating dielectric layer for isolating the first gate dielectric layer from the emitter electrode. A second gate electrode, the heavily doped first conductivity type emitter region and the heavily doped second conductivity type emitter region are connected to one another by means of the emitter electrode. And all front structure units are connected to one another by the emitter electrode.

(8) The first heavily doped first conductivity type emitter region, the second heavily doped first conductivity type emitter region, the lightly doped second conductivity type base region and the first conductivity type drift region form a punch-through triode structure.

(9) The second conductivity type base region, the first conductivity type drift region and the floating FP region form a JFET structure.

(10) Further, the heavily doped first conductivity type emitter region has a doping concentration greater than $1 \times 10^{18} \text{ cm}^{-3}$, and forms an ohmic contact with the emitter electrode. A punch-through voltage of the triode can be changed to 0.3 V to 1 V by adjusting a junction depth and a doping concentration of the lightly doped second conductivity type base region.

(11) Further, the trench auxiliary gate structure includes a second gate electrode and a second gate dielectric layer. The floating FP region is isolated from the second gate electrode by means of the second gate dielectric layer. And the second gate dielectric layer is isolated from the lightly doped second conductivity type base region by means of the second gate dielectric layer.

(12) Further, the second gate dielectric layer is made of silicon oxide, aluminum oxide, and silicon nitride.

(13) Further, the trench auxiliary gate structure includes the second gate dielectric layer and an emitter electrode partially filled in the trench. The second gate dielectric layer is isolated from the lightly doped second conductivity type base region by means of the second gate dielectric layer. And the upper surface of the floating FP region is in contact with the emitter electrode.

(14) The topography of the trench auxiliary gate structure is inverted trapezoidal. The heavily doped first conductivity type emitter region and the lightly doped second conductivity type base region are arranged on a sidewall of the trench auxiliary gate structure, which extends downwards to be in contact with the floating FP region.

(15) The topography of the trench gate structure is inverted trapezoidal.

(16) Further, a doping concentration of the lightly doped second conductivity type base region is

much less than that of each of the second conductivity type base region and the floating FP region, but numerically greater than that of the first conductivity type drift region.

(17) Further, the thickness of the gate dielectric layer is smaller than the width of the gate electrode and is about 0.1 μm to 2 μm . And the gate dielectric layer is made of insulating materials such as silicon dioxide and silicon nitride.

(18) Further, the trench gate structure and the trench auxiliary gate structure have similar depths between 2 μm to 5 μm .

(19) Further, the second conductivity type base region and the floating FP region have the same doping dose and form a Schottky contact with the emitter electrode.

(20) Further, the lateral width of the heavily doped second conductivity type emitter region is smaller than that of the second conductivity type base region. And a ratio of the heavily doped second conductivity type emitter region to the second conductivity type base region is adjustable in the range from 10% to 90%.

(21) Further, the heavily doped second conductivity type emitter region has a depth smaller than 2 μm and is located outside a PN junction depletion region, which is formed by the second conductivity type base region and the first conductivity type drift region.

(22) Further, a N-type channel of the JFET structure formed by the second conductivity type base region, the first conductivity type drift region and the floating FP region has a width between 0.1 μm to 2 μm , such that the JFET channel is completely pinched off when the RC-IGBT is in a blocking state. The JFET channel is formed when the RC-IGBT is forwardly conducted.

(23) Further, the topography of each of the trench gate structure and the trench auxiliary gate structure is inverted trapezoidal. And the sidewalls and bottoms are completely surrounded by the heavily doped first conductivity type emitter region and the floating FP region.

(24) Further, in the same cell, the heavily doped first conductivity type emitter region, the lightly doped second conductivity type base region, the trench auxiliary gate structure and the floating FP region structure are arranged in a staggered manner so as to increase the density of the punch-through NPN triode structure.

(25) Further, the floating FP region has a U-shaped topography and extends upwards to the bottom of the heavily doped first conductivity type emitter region on the upper surface. The topography of the floating FP region is decided by an ion implantation angle and the hot drive-in time.

(26) The present disclosure further provides a fabrication method for the power semiconductor device above. The fabrication method includes the following steps:

(27) Firstly, forming a first conductivity type drift region, and then forming a trench gate structure and a trench auxiliary gate structure on an upper surface of the first conductivity type drift region. Then adjusting the topographies of the trench gate structure and the trench auxiliary gate structure by controlling etching energy, etching time and passivation time.

Secondly, forming a mask layer on the upper surfaces of the trench gate structure and the trench auxiliary gate structure to protect a sidewall of the trench gate structure. Then, forming an ion implantation window, and forming a second conductivity type base region and a floating FP region by means of ion implantation and high temperature drive-in. wherein the floating FP region has a U-shaped topography. The U-shaped topography is obtained by adjusting an ion implantation angle and the thermal drive-in time. Schottky contacts on left and right sidewalls of the trench auxiliary gate structure are achieved by removing gate dielectric layers on the left and right sidewalls of the trench auxiliary gate structure by means of wet etching when opening the contact region.

Thirdly, forming a gate dielectric layer on inner walls of the trench gate structure and the trench auxiliary gate structure, and then forming a gate electrode.

Fourthly, forming a lightly doped second conductivity type base region between the upper surface of second conductivity type base region (adjacent to the trench gate structure) and the trench auxiliary gate structure. And then performing ion implantation to form a heavily doped first conductivity type emitter region, wherein a concentration of the lightly doped second conductivity

type base region is much less than that of the second conductivity type base region. Fifthly, forming an ion implantation window of a heavily doped second conductivity type emitter region on the upper surface of the second conductivity type base region. And then performing ion implantation; and forming an insulating dielectric layer on the upper surface of the device wherein forming a contact hole on the surface of the insulating dielectric layer by means of dry etching, and then forming a front emitter electrode by using a metal sputtering process. Or removing the insulating dielectric layer on the surface of the trench auxiliary gate structure, and then removing the gate electrode and the gate dielectric layer filled in the trench auxiliary gate structure in turn until the upper surface of the floating FP region appears. Forming contact holes on the insulating dielectric layer on the surfaces of the heavily doped first conductivity type emitter region and the heavily doped second conductivity type emitter region by means of dry etching. Then forming a front emitter electrode by using a metal sputtering process, filling the inside of the trench auxiliary gate structure until the front emitter electrode is in connected contact with the upper surface of the floating FP region.

Sixthly, processing the backside of the device to form a first conductivity type buffer layer, a heavily doped second conductivity type collector region, a heavily doped first conductivity type cathode region, and a collector.

(28) Specifically, the method includes: thinning the thickness of the back of the device, and then forming a first conductivity buffer layer. Subsequently, performing local ion implantation in turn on the back of the device by using the mask layer to form a heavily doped second conductivity type collector region and a heavily doped first conductivity type cathode region, which is arranged in a staggered manner and to be a collector on the backside of the device.

(29) Compared with the prior art, the present disclosure has the beneficial effects that: (1) By introducing the punch-through NPN triode to extract front electron currents, the hole injection efficiency of the emitter region when the RC-IGBT is in free-wheeling can be adjusted. Compared with the barrier height adjustment of the Schottky junction and an individual PN junction, the barrier height of the punch-through NPN triode is easy to adjust, and the reduction of reverse recovery loss is easier to achieve. (2) The barrier height for frontside electron extraction consists of a PN junction depletion region, while there is a JFET structure for shielding a high electric field, which can effectively reduce the leakage current at the high electric field. (3) The reverse recovery loss reduction effect close to that of the local lifetime control technology can be achieved. And the leakage current, process difficulty and fabrication cost at high temperatures are reduced. (4) The adjustment of the area and position of punch-through NPN is achieved by changing the density and topography of the frontside trench of the RC-IGBT, which in turn can change the electron extraction and frontside hole injection efficiency. This method increases the design flexibility and design dimension of the device. (5) The structure is simultaneously suitable for IGBT cells and the FWD cell in the RC-IGBT. It is no need of individual partition design and it can reduce the reverse recovery loss of the RC-IGBT without degrading the characteristics of the RC-IGBT in an IGBT operation mode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of a cross section of a RC-IGBT device structure with a trench auxiliary gate in the prior art.

(2) FIG. 2 is a schematic diagram of a cross section of a RC-IGBT device structure in accordance with a first embodiment of the present disclosure.

(3) FIG. 3 to FIG. 8 are schematic diagrams of cross sections of main fabrication steps in accordance with a first embodiment of the present disclosure.

(4) FIG. 9 is a schematic diagram of a cross section of a RC-IGBT device structure in accordance with a second embodiment of the present disclosure.

(5) FIG. 10 to FIG. 17 are schematic diagrams of cross sections of main fabrication steps in accordance with a second embodiment of the present disclosure.

(6) FIG. 18 is a schematic diagram of a cross section of a RC-IGBT device structure in accordance with a third embodiment of the present disclosure.

(7) FIG. 19 is a schematic diagram of a cross section of a RC-IGBT device structure in accordance with a fourth embodiment of the present disclosure.

DESCRIPTION OF THE EMBODIMENTS

(8) The present disclosure provides a RC-IGBT device and a preparation method thereof, and provides the following specific embodiments.

(9) It needs to be noted that in the description of the following embodiments, it is considered that a semiconductor substrate of the RC-IGBT is made of a Si material. However, the substrate may also be made of any other materials suitable for the preparation of the device, such as silicon carbide (SiC) and germanium (Ge).

(10) It needs to be noted that corresponding positional words described herein, such as “upper,” “lower,” “left,” “right,” “front,” and “rear,” are relative positions corresponding to those illustrated with reference to drawings. The fixed direction is not limited in the specific implementation, and the frontside structures of the device may consist of a plurality of repeatable units. The following is a description of the minimum front structural unit. In practical implementation, the IGBT cell portion in the frontside structure of the device may be replaced with cell structures of the discrete IGBT in the prior art. The device structure is not limited to silicon-based RC-IGBTs and is also applicable for silicon-based field effect transistors (MOSFET), silicon carbide-based MOSFETs, and silicon carbide-based RC-IGBTs.

Embodiment 1

(11) FIG. 2 is a schematic diagram of a cross section of a RC-IGBT device in accordance with a first embodiment of the present disclosure. The device structure includes: a collector electrode **201** located at the bottom, a heavily doped N.sup.+ cathode region **202** and a heavily doped P.sup.+ collector region **203** which are located on the collector region **201** in a staggered manner. A N-type buffer layer **204** is located on the heavily doped N.sup.+ cathode region **202** and the heavily doped P.sup.+ collector region **203**. And a N-type drift region **205** is located on the N-type buffer layer **204**. An upper surface of the N-type drift region **205** consists of a plurality of repeating type frontside structures. The minimum front structural unit includes: a trench gate structure **206**, a P-type base region **207**, a trench auxiliary gate structure **208**, a lightly doped P.sup.-type base region **215**, and a floating FP region **216**. The trench gate structure **206** is sequentially filled with a first gate dielectric layer **209** and a first gate electrode **210**. The P-type base region **207** is adjacent to the trench gate structure **206** and is isolated from the first gate electrode **210** by means of the first gate dielectric layer **209**. The trench auxiliary gate structure **208** is sequentially formed by a second gate dielectric layer **217** and a second gate electrode **218**. And it is isolated from the P-type base region **207** by means of a lightly doped P.sup.-type base region **215**. The floating FP region **216** is located at the bottom of the trench auxiliary gate structure **208**. And it is isolated from the second gate electrode **218** in the trench auxiliary gate structure **208** by means of the second gate dielectric layer **217** which is isolated from the P-type base region **207** and the lightly doped P.sup.-type base region **215** by means of the N-type drift region **205**. An upper surfaces of the P-type base region **207** and the lightly doped P.sup.-type base region **215** are both provided with heavily doped N.sup.+ -type emitter regions **211**. The heavily doped N.sup.+ -type emitter regions **211** are separated by a heavily doped P.sup.+ -type emitter region **212** on the upper surface of the P-type base region **207**. An upper surface of the trench gate structure **206** is provided with an insulating dielectric layer **213**. The insulating dielectric layer **213** is configured to isolate an emitter electrode **214** from the gate electrode **210** in the trench gate structure **206**. A second gate electrode **210** in the

trench auxiliary gate structure **208**, the heavily doped N.sup.+ -type emitter regions **211** and the heavily doped P.sup.+ -type emitter region **212** are connected to one another by means of the emitter electrode **214**. The plurality of repeating type frontside structures are connected to one another by means of the emitter electrode **214**. Particularly, the heavily doped N.sup.+ -type emitter region **211**, the lightly doped P.sup.- -type base region **215** and the N-type drift region **205** form a punch-through NPN triode structure. The P-type base region **207**, the N-type drift region **205** and the floating FP region **216** form a JFET structure.

(12) The heavily doped N.sup.+ -type emitter region **211** has a doping concentration greater than $1 \times 10^{18} \text{ cm}^{-3}$ and forms an ohmic contact with the emitter electrode **214**. A punch-through voltage of the NPN triode is changed to 0.3 V to 1 V by adjusting a junction depth and a doping concentration of the lightly doped P.sup.- -type base region **215**.

(13) The doping concentration of the lightly doped P.sup.- -type base region **215** is much less than that of each of the P-type base region **207** and the floating FP region **216**, but numerically greater than that of the N-type drift region **205**. The thickness of the gate dielectric layer **209** is smaller than the width of the gate electrode **210** and is about 0.1 μm to 2 μm . The gate dielectric layer is made of insulating materials such as silicon dioxide and silicon nitride. The trench gate structure **206** and the trench auxiliary gate structure **208** have similar depths between 2 μm to 5 μm .

(14) The P-type base region **207** and the floating FP region **216** have the same doping dose and form a Schottky contact with the emitter electrode **214**. The heavily doped Pt-type emitter region **212** has a doping concentration greater than $1 \times 10^{18} \text{ cm}^{-3}$ and forms an ohmic contact with the emitter electrode **214**. A lateral width of the heavily doped Pt-type emitter region **212** is less than that of the P-type base region **207**. A ratio of the heavily doped second conductivity type emitter region to the second conductivity type base region is adjustable in the range from 10% to 90%. Further, the heavily doped Pt-type emitter region **212** has a depth less than 2 μm and is located outside a PN junction depletion region formed by the P-type base region **207** and the N-type drift region **205**. A N-type channel of the JFET structure formed by P-type base region **207**, the N-type drift region **205** and the floating FP region **216** has a width between 0.1 μm to 2 μm , such that the JFET channel is completely pinched off when the RC-IGBT is in a blocking state. The JFET channel is formed when the RC-IGBT is forwardly conducted.

(15) In the same cell, the heavily doped N.sup.+ -type emitter region **211**, the lightly doped P.sup.- -type base region **215**, the trench auxiliary gate structure **208** and the floating FP region structure **216** are arranged in a staggered manner so as to increase the density of the punch-through NPN triode structure.

(16) In addition, the present disclosure further provides a fabrication method for the device of the first embodiment, as shown in FIGS. 3-8:

(17) Firstly, a N-type drift region **205** is formed. A mask layer is formed on an upper surface of the N-type drift region **205**. And a vertical type trench gate structure **206** and a trench auxiliary gate structure **208** are formed by means of dry etching, wherein the trench has a depth of 2 μm to 5 μm , as shown in FIG. 3.

(18) Secondly, as shown in FIG. 4, a mask layer is formed on the upper surfaces of the trench gate structure **206** and the trench auxiliary gate structure **208** to protect a sidewall of the trench gate structure. Meanwhile, an ion implantation window is formed, and a P-type base region **207** and a floating FP region **216** are formed by means of ion implantation and high-temperature drive-in. A commonly used impurity for P-type implantation is boron.

(19) Thirdly, as shown in FIG. 5, a gate dielectric layer, which may be made of silicon oxide, aluminum oxide and silicon nitride, is formed on the inner walls of the trench gate structure **206** and the trench auxiliary gate structure **208** by using a high temperature thermal growth or chemical vapor deposition process. Then a gate electrode, which may be made of heavily doped polycrystalline silicon, is formed by using the chemical vapor deposition process.

(20) Fourthly, as shown in FIG. 6, an ion implantation window of a punch-through NPN triode is

formed between the P-type base region **207** and the trench auxiliary gate structure **208** as well as the upper surface of the P-type base region **207** adjacent to the trench gate structure **206**. The P-type ion implantation and high-temperature drive-in process are performed firstly to form a lightly doped P.sup.--type base region **215**. A commonly used impurity for P-type implantation is boron. Then N-type ion implantation is performed to form a heavily doped N.sup.+type emitter region **211**. A commonly used impurity for N-type implantation is arsenic. The concentration of the lightly doped P.sup.--type base region **215** is much less than that of the P-type base region **207**. Then, the change in the doping concentrations of the P-type base region **207** and the heavily doped N.sup.+type emitter region **211** is negligible.

(21) Fifthly, as shown in FIG. 7, an ion implantation window of a heavily doped P.sup.+type emitter region **212** is formed on the upper surface of the P-type base region **207** by dry-etching the mask layer, and then P-type ion implantation is performed. A commonly used impurity for P-type implantation is boron. An insulating dielectric layer **213** is formed on the upper surface of the device by using a high temperature thermal growth or chemical vapor deposition process. A contact hole is formed on the surface of the insulating dielectric layer **213** by means of dry etching, and then a front emitter electrode **214** is formed by using a metal sputtering process.

(22) Finally, as shown in FIG. 8, the back thickness of the device is thinned by using a thinning process. Then a N-type buffer layer **204** is formed by a N-type impurity diffusion or ion implantation process. Subsequently, local P-type ion implantation and local N-type ion implantation are performed in sequence on the back of the device by using the mask layer to form a heavily doped P.sup.+ collector region **203** and a heavily doped $1\backslash 1+$ cathode region **202**, which is arranged in a staggered manner. Finally, a backside collector electrode **201** is formed by using the metal sputtering process.

Embodiment 2

(23) FIG. 9 is a schematic diagram of a cross section of a RC-IGBT device in accordance with a second embodiment of the present disclosure. Compared with the device structure in the first embodiment shown in FIG. 2, the difference of the device structure in the second embodiment is that an upper surface of a floating FP region **316** is directly connected to an emitter electrode **314** at the bottom of a trench auxiliary gate structure **308**. While in the first embodiment of FIG. 2, a Schottky contact is formed between the floating FP region **216** and the emitter electrode **214**. In the second embodiment, such design increases the area and hole pathways of the frontside emitter region of the RC-IGBT. And the Schottky contact may control the hole injection efficiency. This design is conducive to reducing on-state voltage drop and reverse recovery loss of the device.

(24) The device structure consists of: a collector electrode **301** located at the bottom, a heavily doped N.sup.+ cathode region **302** and a heavily doped P.sup.+ collector region **303** located on the collector region **301** in a staggered manner, a N-type buffer layer **304** located on the heavily doped N.sup.+ cathode region **302** and the heavily doped P.sup.+ collector region **303**, and a N-type drift region **305** located on the N-type buffer layer **304**. An upper surface of the N-type drift region **305** consists of a plurality of repeating type frontside structures. The minimum front structural unit includes: a trench gate structure **306**, a P-type base region **307**, a trench auxiliary gate structure **308**, a lightly doped P.sup.--type base region **315**, and a floating FP region **316**. The trench gate structure **306** is sequentially filled with a first gate dielectric layer **309** and a first gate electrode **310**. The P-type base region **307** is adjacent to the trench gate structure **306** and is isolated from the first gate electrode **310** by means of the gate dielectric layer **309**. The trench auxiliary gate structure **308** is sequentially formed by a second gate dielectric layer **317** and an emitter electrode **318** partially filled in the trench, which is isolated from the P-type base region **307** by means of a lightly doped P.sup.--type base region **315**. An emitter electrode **314** in the trench auxiliary gate structure **308** is isolated from the lightly doped P.sup.--type base region **315** by means of the second gate dielectric layer **309**. The floating FP region **316** is located at the bottom of the trench auxiliary gate structure **308** and is directly connected to the emitter electrode **314** at the bottom of

the trench auxiliary gate structure **308**. The floating FP region is also isolated from the emitter electrode **314** on the sidewall of the trench auxiliary gate structure **308** by means of the gate dielectric layer **309**. The floating FP region **316** is separated from the P-type base region **307** and the lightly doped P.sup.-type base region **315** by means of the N-type drift region **305**. Upper surfaces of the P-type base region **307** and the lightly doped P.sup.-type base region **315** are both provided with heavily doped N.sup.+type emitter regions **311**. The heavily doped N.sup.+type emitter regions **311** are separated by a heavily doped P-type emitter region **312** on the upper surface of the P-type base region **307**. An upper surface of the trench gate structure **306** is provided with an insulating dielectric layer **313**. And the insulating dielectric layer **313** is configured to isolate the emitter electrode **314** from the first gate electrode **310** in the trench gate structure **306**. The emitter electrode **314** in the trench auxiliary gate structure **308**, the heavily doped N.sup.+type emitter region **311** and the heavily doped P.sup.+type emitter region **312** are connected to one another by means of the emitter electrodes **314** on the upper surface of the device. The plurality of repeating type frontside structures are connected to one another by means of the emitter electrodes **314**. Particularly, the heavily doped N.sup.+type emitter region **311**, the lightly doped P.sup.-type base region **315** and the N-type drift region **305** form a punch-through NPN triode structure. The P-type base region **307**, the N-type drift region **305** and the floating FP region **316** form a JFET structure.

(25) The main operating principle and other key structural parameters of the device are consistent with those of the device in the first embodiment. As such, necessary details will not be repeated here.

(26) In addition, the present disclosure further provides a fabrication method for the device of the second embodiment, as shown in FIGS. **10-17**.

(27) Firstly, a N-type drift region **305** is formed. A mask layer is formed on an upper surface of N-type drift region **305**. And a vertical type trench gate structure **306** and a trench auxiliary gate structure **308** are formed by means of dry etching, where the trench has a depth of 2 μm to 5 μm , as shown in FIG. **10**.

(28) Secondly, as shown in FIG. **11**, mask layer is formed on the upper surfaces of the trench gate structure **306** and the trench auxiliary gate structure **308**, which can protect a sidewall of the trench gate structure. Meanwhile, an ion implantation window is formed. A P-type base region **307** and a floating FP region **316** are formed by means of ion implantation and high-temperature drive-in process. A commonly used impurity for P-type implantation is boron.

(29) Thirdly, as shown in FIG. **12**, a gate dielectric layer, which may be made of silicon oxide, aluminum oxide and silicon nitride, is formed on the inner walls of the trench gate structure **306** and the trench auxiliary gate structure **308** by using a high temperature thermal growth or chemical vapor deposition process. Then a second gate dielectric layer **317** and an emitter electrode **318** partially filled in the trench are formed by the chemical vapor deposition process, wherein the gate electrode may be made of heavily doped polycrystalline silicon.

(30) Fourthly, as shown in FIG. **13**, an ion implantation window of a punch-through NPN triode is formed between the P-type base region **307** and the trench auxiliary gate structure **308** as well as the upper surface of the P-type base region **307** adjacent to the trench gate structure **306**. The P-type ion implantation and high-temperature drive-in are performed firstly to form a lightly doped P.sup.-type base region **315**. A commonly used impurity for P-type implantation is boron. Then N-type ion implantation is performed to form a heavily doped N.sup.+type emitter region **311**. A commonly used impurity for N-type implantation is arsenic. The concentration of the lightly doped P.sup.-type base region **315** is much less than that of the P-type base region **307**. Therefore, the change in the doping concentrations of the P-type base region **307** and the heavily doped N.sup.+type emitter region **311** is negligible.

(31) Fifthly, as shown in FIG. **14**, an ion implantation window of the heavily doped P.sup.+type emitter region **312** is formed on the upper surface of the P-type base region **307** by dry-etching the

mask layer. Then the P-type ion implantation is performed. A commonly used impurity for P-type implantation ion is boron. Then, an insulating dielectric layer **313** is formed on the upper surface of the device by using a high temperature thermal growth or chemical vapor deposition process.

(32) Sixthly, as shown in FIG. **15**, the insulating dielectric layer **313** on the surface of the trench auxiliary gate structure **308** is removed by means of dry etching. Then the second gate dielectric layer **317** filled in the trench auxiliary gate structure **308** and the emitter electrode **318** partially filled in the trench are removed successively by means of dry etching or wet etching until an upper surface of the floating FP region **316** appears.

(33) Seventhly, as shown in FIG. **16**, a contact hole is formed on the insulating dielectric layer **313** on the surfaces of the heavily doped N.sup.+ -type emitter region **311** and the heavily doped Pt-type emitter region **312** by means of dry etching. Then a frontside emitter electrode **314** is formed by using a metal sputtering process. The frontside emitter electrode is configured to fill the inside of the trench auxiliary gate structure **308** until the front emitter electrode **314** is in interconnected contact with the upper surface of the floating FP region **316**.

(34) Finally, as shown in FIG. **17**, the backside thickness of the device is thinned by using a thinning process. Then a N-type buffer layer **304** is formed by a N-type impurity diffusion or ion implantation process. Subsequently, a local P-type ion implantation and local N-type ion implantation are performed in sequence on the backside of the device by using the mask layer to form a heavily doped P.sup.+ collector region **303** and a heavily doped N.sup.+ cathode region **302** arranged in a staggered manner. Finally, a backside collector electrode **301** is formed by using the metal sputtering process.

Embodiment 3

(35) FIG. **18** is a schematic diagram of a cross section of a cell structure of a RC-IGB device in accordance with a third embodiment of the present disclosure. Compared with the device structure of the second embodiment of the present disclosure, the device structure of the third embodiment further has the following features: the topography of each of a trench gate structure **406** and a trench auxiliary gate structure **408** is inverted trapezoidal. A lightly doped P.sup.- -type base region **415** and a heavily doped N.sup.+ -type emitter region **411** on the surface thereof are both located on the left and right sidewalls of a trench auxiliary gate structure **408**. Such design increases the area of the left and right sidewalls of the trench auxiliary gate structure as the frontside emitter region as well as the conductive area. Therefore, it is conducive to improving an area ratio of the punch-through NPN triode structure on the front, which can enhance the extraction of electrons by the frontside FWD cell and reduce the reverse recovery loss. For the fabrication method, the topography control of the trench gate structure **406** and the trench auxiliary gate structure **408** is achieved by controlling etching energy, etching time and passivation time by using a Bosch process. The punch-through NPN triode structure located on the left and right sidewalls of the trench auxiliary gate structure **408** can be achieved by removing the gate dielectric layer on the left and right sidewalls of the trench auxiliary gate structure **408** using wet etching when opening the contact region.

Embodiment 4

(36) FIG. **19** is a schematic diagram of a cross section of cell structures of a RC-IGBT device in accordance with a fourth embodiment of the present disclosure. Compared with the device structure of the second embodiment of the present disclosure, the device structure of the fourth embodiment further has the following features: a floating FP region **516** has a U-shaped topography. The left and right sidewalls and the bottom of a trench auxiliary gate structure **508** are completely surrounded and are all connected with an emitter electrode **514** to form Schottky contacts. And an electron channel of a JFET structure formed by a P-type base region **507**, a N-type drift region **505** and the floating FP region **516** is narrower. Such design increases the shielding effect of the JFET structure against high electric fields when the RC-IGBT is in a blocking state, which is conducive to reducing the leakage current. Meanwhile, the left and right sidewalls of the trench auxiliary gate

structure **508** are conductive area of the emitter region, especially the conductive area of the hole, which is conducive to reducing the voltage drop when the RC-IGBT is forwardly conducted.

(37) For the fabrication method, the U-shaped topography of the floating FP region **516** is controlled by an ion implantation angle and thermal drive-in time. The Schottky contacts on the left and right sidewalls of the trench auxiliary gate structure **508** are achieved by removing the gate dielectric layer **509** on the left and right sidewalls of the trench auxiliary gate structure **508** using wet etching when opening the contact region.

(38) The operating principle of the semiconductor device of the present disclosure is explained as follows:

(39) Taking the RC-IGBT as an example, when the RC-IGBT is conducted for freewheeling, i.e., a gate voltage is less than zero, an emitter-to-collector voltage V_{ec} is higher than on-state voltage drop, FWD cell in the RC-IGBT is in a small current-on state. There is hole injection in the front of the IGBT cell, and corresponding back is almost non-conductive. In accordance with an existing RC-IGBT device structure shown in FIG. 1, to guarantee the breakdown voltage of the IGBT cell portion and suppress the latch-up effect, the P region in the frontside structure is typically heavily doped and the hole injection efficiency at the moment is too high, resulting in increased loss of the FWD cell in subsequent reverse recovery process.

(40) The punch-through NPN triode of the proposed structure has a classical punch-through voltage drop value of 0.7 V. The hole injection in the frontside structure at the moment mainly involves the heavily doped Pt-type emitter region, the P-type base region and the floating FP region. Further, when the floating FP region is in contact with the emitter electrode, the floating FP region and the P-type base region may both form Schottky contact with the emitter electrode due to work function difference of materials. The polarity characteristics of the Schottky junction make the voltage drop be partially distributed to both ends of the Schottky junction, such that the potential of the P-type base region and the floating FP region is reduced. Then, in turn the hole injection efficiency decreases. The formed Schottky junction has no effect on the diffusion of electrons from the P-type region to the emitter electrode. Compared with the FWD cell structure in the existing RC-IGBT, the punch-through NPN triode structure for electron extraction in the drift region is introduced into the proposed structure, thus the frontside hole injection efficiency can be reduced. And hole currents of emitters can be converted into electron drift currents without significantly increasing on-state voltage drop. At the punch through state of the punch-through NPN triode in the proposed structure, the electric field intensity of the lightly doped P.sup.-type base region of the NPN is greater than $1e4$ V/cm, where electrons are close to reaching velocity saturation. The carrier velocity overshoot may occur when the junction depth of the lightly doped P-type base region is shallow. At the moment, the electrons close to the frontside position of the drift region are transported to the emitter electrode at a saturation velocity due to high electric fields of the lightly doped P.sup.-type base region of the NPN. Therefore, more hole currents from emitters are converted into the electron drift currents. Then, the hole injection and the number of carriers in the drift region are reduced. In turn, when the device is conducted at low currents, the carrier distribution are low in the front and high in the back across the drift region.

(41) When the FWD cells of the proposed structure enter the reverse recovery, i.e., the gate voltage is less than zero, and the emitter-to-collector voltage V_{ec} gradually decreases to zero and reverse conduction is started. The carriers in the drift region has the distribution profile that the frontside carrier concentration is low while the overall number is reduced. Then the reverse recovery charge is reduced and a PN junction depletion region in the drift region can be rapidly broadened to sweep out the carriers to withstand the blocking voltage, which reduces the reverse recovery time. The carriers in the drift region have the distribution profile that the backside carrier concentration is high. Then the current may have enough carriers to achieve free-wheeling at the tail current stage, which can avoid the sudden decrease of reverse recovery current and suppress the generation of EMI noise.

(42) When the RC-IGBT device is in a blocking state, i.e., the gate voltage is less than zero, and the emitter-to-collector voltage V_{ec} is less than zero. The JFET structure formed by the P-type base region and the floating FP region with the N-type drift region, or the adjacent floating FP regions with the N-type drift region, may pinch off the JFET channel region. Then the effect of the high electric fields on the punch-through NPN triode is shielded. Meanwhile, during punch through, the depletion region is a PN junction, which is more insensitive to the high electric fields and high temperatures and cannot increase the leakage current significantly. The proposed RC-IGBT structure utilizes the trench auxiliary gate structure in the prior art. It also introduces the floating FP region and the punch-through NPN triode structure, which is compatible with an existing RC-IGBT process without affecting the switching process of the IGBT cell portion of the RC-IGBT.

(43) Further, when the position of the punch-through NPN triode is adjusted to the sidewall of the inverted-trapezoidal trench auxiliary gate structure, or the floating FP region has a U-shaped topography, the conductive area of the emitter region and the electron extraction capacity of the punch-through NPN triode are effectively increased, which is conducive to reducing the on-state voltage drop and improving the forward surge current capacity of the device. With the increase of the density of the trench gate, the punch-through NPN triode has an increased area for frontside electron extraction, which is more conducive to reducing frontside carrier distribution in the drift region and further reducing the reverse recovery loss of the RC-IGBT.

Claims

1. A power semiconductor device, comprising a collector located at the bottom, a heavily doped first conductivity type cathode region and a heavily doped second conductivity type collector region arranged on the collector in a staggered manner, a first conductivity type buffer layer located on the heavily doped first conductivity type cathode region and a heavily doped collector region, a first conductivity type drift region located on the first conductivity type buffer layer, and an emitter electrode located on the top of the device, the emitter electrode is separated from the first conductivity type drift region, wherein more than one frontside structure units are arranged on the first conductivity type drift region, each of the frontside structure units comprises: a trench gate structure, a second conductivity type base region adjacent to the trench gate structure, a lightly doped second conductivity type base region located on one side of the second conductivity type base region to isolate a trench auxiliary gate structure from the second conductivity type base region, the trench auxiliary gate structure, and a floating FP region located at the bottom of the trench auxiliary gate structure; the trench gate structure comprises a first gate dielectric layer and a first gate electrode; the floating FP region is separated from the second conductivity type base region and the lightly doped second conductivity type base region; an upper surface of the second conductivity type base region is provided with a first heavily doped first conductivity type emitter region and a heavily doped second conductivity type emitter region, an upper surface of the lightly doped second conductivity type base region is provided with a second heavily doped first conductivity type emitter region, and the first heavily doped first conductivity type emitter region is separated from the second heavily doped first conductivity type emitter region; an upper surface of the trench gate structure is provided with an insulated dielectric layer for isolating the first gate dielectric layer from the emitter electrode, a second gate electrode is arranged in the trench auxiliary gate structure, the second gate electrode, the heavily doped first conductivity type emitter region and the heavily doped second conductivity type emitter region are connected to one another by means of the emitter electrode, and all frontside structure units are connected to one another by the emitter electrode; the second heavily doped first conductivity type emitter region, the lightly doped second conductivity type base region and the first conductivity type drift region form a punch-through triode structure, the second conductivity type base region, the first conductivity type drift region and the floating FP region form a JFET structure; the second conductivity type base

region and the floating FP region have the same doping dose, and the second conductivity type base region and the floating FP region form a Schottky contact with the emitter electrode; the second gate electrode is formed by the emitter electrode filled in the trench, and an upper surface of the floating FP region is directly connected to the emitter electrode at the bottom of the trench auxiliary gate structure; and the topography of the trench gate structure and the trench auxiliary gate structure is inverted trapezoidal, the lightly doped second conductivity type base region and the second heavily doped first conductivity type emitter region on the surface thereof are both located on left and right sidewalls of the trench auxiliary gate structure.

2. The power semiconductor device according to claim 1, wherein the heavily doped first conductivity type emitter region has a doping concentration greater than $1 \times 10^{18} \text{ cm}^{-3}$, and it forms an ohmic contact with the emitter electrode; and by adjusting a junction depth and a doping concentration of the lightly doped second conductivity type base region, a punch-through voltage of the triode structure is changed to be 0.3 V to 1 V.

3. A fabrication method for the power semiconductor device according to claim 1, comprising the following steps: firstly, forming a first conductivity type drift region, and forming a trench gate structure and a trench auxiliary gate structure on an upper surface of the first conductivity type drift region; secondly, forming a mask layer on the surfaces of the trench gate structure and the trench auxiliary gate structure so as to protect a sidewall of the trench gate structure, forming an ion implantation window at the same time, and then forming a second conductivity type base region and a floating FP region by means of ion implantation and thermal drive-in, the temperature for high temperature drive-in is 900°C. to 1200°C. ; thirdly, forming a gate dielectric layer on inner walls of the trench gate structure and the trench auxiliary gate structure, and then forming gate electrodes; fourthly, forming an ion implantation window of a punch-through triode between the second conductivity type base region and the trench auxiliary gate structure as well as the upper surface of the second conductivity type base region adjacent to the trench gate structure; performing second conductivity type ion implantation to form a lightly doped second conductivity type base region, and then performing first conductivity type ion implantation to form a heavily doped first conductivity type emitter region; fifthly, forming an ion implantation window of a heavily doped second conductivity type emitter region on the upper surface of the second conductivity type base region, and then performing ion implantation; forming an insulated dielectric layer, and finally forming a frontside emitter electrode on the upper surface of the device; sixthly, processing the back of the device to form a first conductivity type buffer layer, a heavily doped second conductivity type collector region, a heavily doped first conductivity type cathode region, and a collector.

4. The fabrication method for the power semiconductor device according to claim 3, wherein in the step one, the topography of the trench gate structure and the trench auxiliary gate structure is adjusted by controlling etching energy, etching time, and passivation time.
